

INFORMATION TECHNOLOGY INSTITUTE

Analog IC Design Track



Analog Integrated Circuit Design

Lab 10

Noise Simulation

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PART 1: LPF AC Noise Analysis

1.1 Output Noise vs. Frequency

1.1.1 Simulation Results:



Figure 1: output voltage noise spectral density vs frequency for LPF ($R = 1 \text{ k}\Omega$, $C = 1 \text{ pF}$)

As we see

Low-Frequency Voltage Noise Density: $4.07 \text{ nV}/\sqrt{\text{Hz}}$

Bandwidth: 159.15 MHz

1.1.2 Hand Analysis:

Thermal Noise Density:

The input-referred thermal noise generated by the resistor is given by:

$$v_n = \sqrt{4kTR}$$

$$v_n = \sqrt{4 \times (1.38 \times 10^{-23}) \times 300 \times 1000} \approx 4.07 \text{ nV}/\sqrt{\text{Hz}}$$

Bandwidth:

$$f_{3dB} = \frac{1}{2\pi RC}$$

$$f_{3dB} = \frac{1}{2\pi(1000)(10^{-12})} \approx 159.15 \text{ MHz}$$

Total RMS Noise ($v_{n,rms}$):

$$v_{n,rms} = \sqrt{\frac{kT}{C}}$$

$$v_{n,rms} = \sqrt{\frac{1.38 \times 10^{-23} \times 300}{10^{-12}}} \approx 64.34 \mu V$$

ITI_LABS:lab_10_noise_rc:1	output noise; V / sqrt(Hz)	
ITI_LABS:lab_10_noise_rc:1	rmsNoise(1 1e+10)	64.32u

Figure 2: total RMS noise

Table 1: Comparison Table:

Parameter	Hand Calc.	Simulation
Noise Density	4.07 nV/ $\sqrt{\text{Hz}}$	4.071 nV/ $\sqrt{\text{Hz}}$
BW	159.15 MHz	159.15 MHz
Total RMS	64.34 μV	64.32 μV

1.2 Parametric Sweep

1.2.1 Simulation Results:

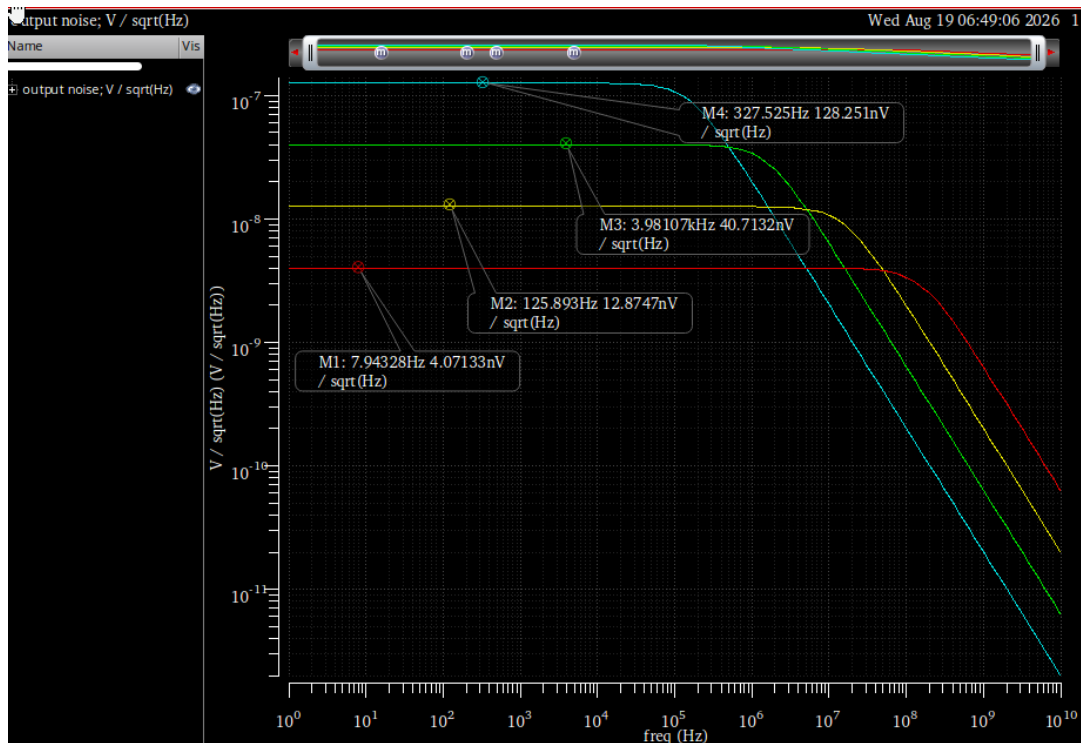


Figure 3: Overlaid AC output noise spectral density plots for $R = 1k\Omega, 10k\Omega, 100k\Omega$, and $1M\Omega$

$$R = 1 \text{ k}\Omega: \approx 4.07 \text{ nV}/\sqrt{\text{Hz}}$$

$$R = 10 \text{ k}\Omega: \approx 12.87 \text{ nV}/\sqrt{\text{Hz}}$$

$$R = 100 \text{ k}\Omega: \approx 40.71 \text{ nV}/\sqrt{\text{Hz}}$$

$$R = 1 \text{ M}\Omega: \approx 128.25 \text{ nV}/\sqrt{\text{Hz}}$$

Parameters: R=1k			
1	ITI_LABS:lab_10_noise_rc:1	output noise; V / sqrt(Hz)	
1	ITI_LABS:lab_10_noise_rc:1	rmsNoise(1 1e+10)	64.32u
Parameters: R=10k			
2	ITI_LABS:lab_10_noise_rc:1	output noise; V / sqrt(Hz)	
2	ITI_LABS:lab_10_noise_rc:1	rmsNoise(1 1e+10)	64.62u
Parameters: R=100k			
3	ITI_LABS:lab_10_noise_rc:1	output noise; V / sqrt(Hz)	
3	ITI_LABS:lab_10_noise_rc:1	rmsNoise(1 1e+10)	64.65u
Parameters: R=1M			
4	ITI_LABS:lab_10_noise_rc:1	output noise; V / sqrt(Hz)	
4	ITI_LABS:lab_10_noise_rc:1	rmsNoise(1 1e+10)	64.66u

Figure 4: total RMS output noise for varying R

the total thermal noise of a simple RC circuit is independent of the resistance. Although a larger resistor generates a higher thermal noise spectral density ($\propto \sqrt{R}$), it proportionally restricts the noise bandwidth ($\propto 1/R$). During integration, the resistance terms cancel out, leaving the total noise voltage entirely constrained by the load capacitance ($v_{n,rms} = \sqrt{kT/C}$).

2 PART 2: LPF Transient Noise Analysis

2.1 Transient Noise Analysis

2.1.1 Simulation Results:

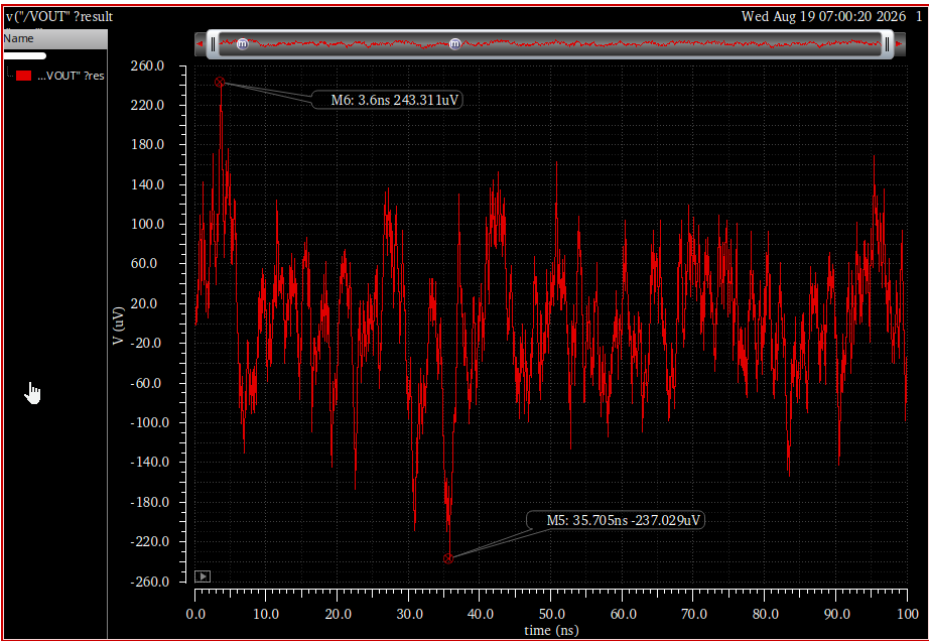


Figure 5: Transient noise output waveform

From the simulation plot:

- Maximum Value: 243.31 μV at 3.6 ns
- Minimum Value: $-237.03 \mu\text{V}$ at 35.7 ns

2.1.2 RMS Noise Comparison

ITI_LABS:lab_10_noise_rc:1	v("/VOUT" ?result "tran")	
ITI_LABS:lab_10_noise_rc:1	rms(VT("/VOUT"))	63.53u

Figure 6: RMS calculation of the output noise

This value (63.53 μV) is close to the AC noise RMS value obtained in Part 1 (64.32 μV).

2.1.3 Effect of Time Step (TSTEP) on Noise:

The simulation was repeated with a larger time step, changing TSTEP from TAU/100 to TAU/10.

ITI_LABS:lab_10_noise_rc:1	v("/VOUT" ?result "tran")	
ITI_LABS:lab_10_noise_rc:1	rms(VT("/VOUT"))	66.57u

Figure 7: RMS calculation of the output noise with $TSTEP = TAU/10$.

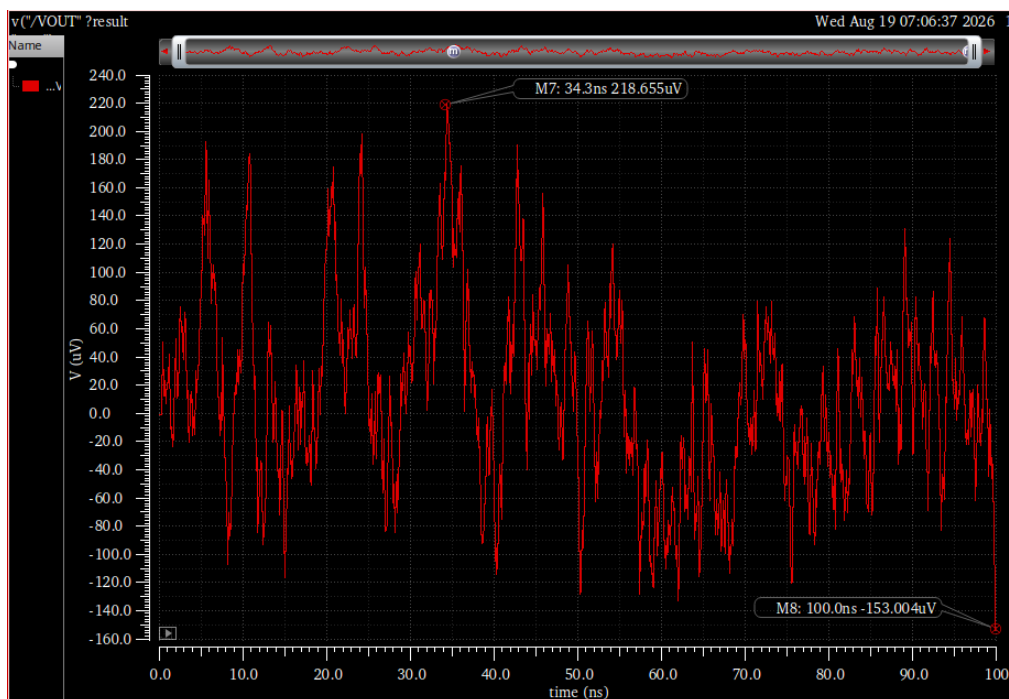


Figure 8: Transient noise output waveform with ($TSTEP = TAU/10$).

Output Noise RMS Increased Slightly as Fmax increased so the area integrated is bigger.

2.1.4 20 Iterations Analysis

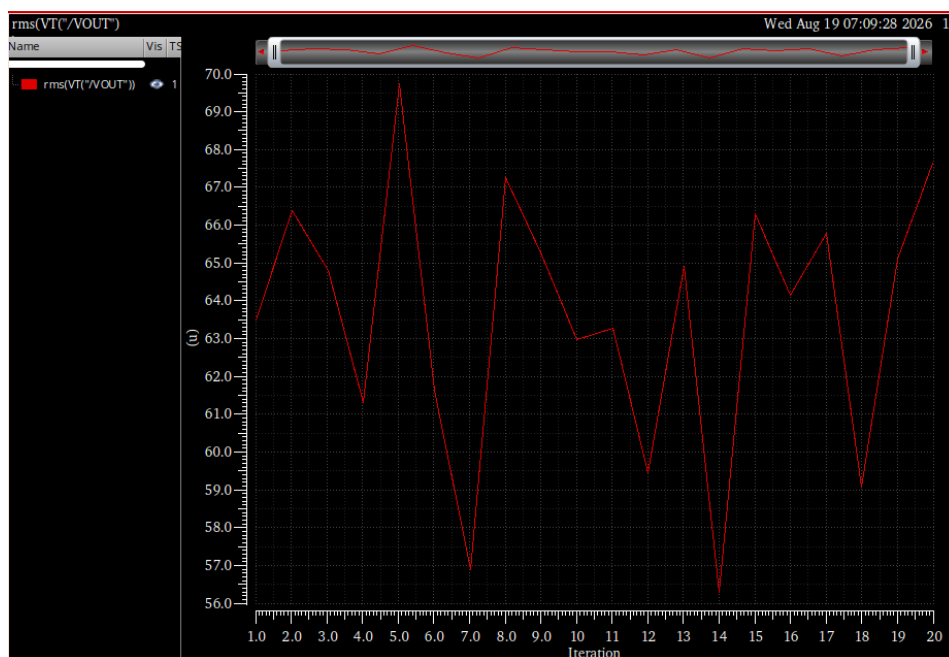


Figure 9: RMS noise versus iteration number for 20 distinct transient noise simulation

ITI_LABS:lab_10_noise_rc:1	v("/VOUT" ?result "tran")	
ITI_LABS:lab_10_noise_rc:1	rms(VT("/VOUT"))	
ITI_LABS:lab_10_noise_rc:1	average(rms(VT("/VOUT")))	63.51u

Figure 1: Average RMS Noise

The averaged transient RMS noise ($63.51 \mu\text{V}$) is very closely to the AC simulation result from Part 1 ($64.32 \mu\text{V}$), although individual transient noise simulations are subject to high random variance.

3 PART 3: 5T OTA AC Noise Analysis

3.1 Thermal Noise vs. Frequency

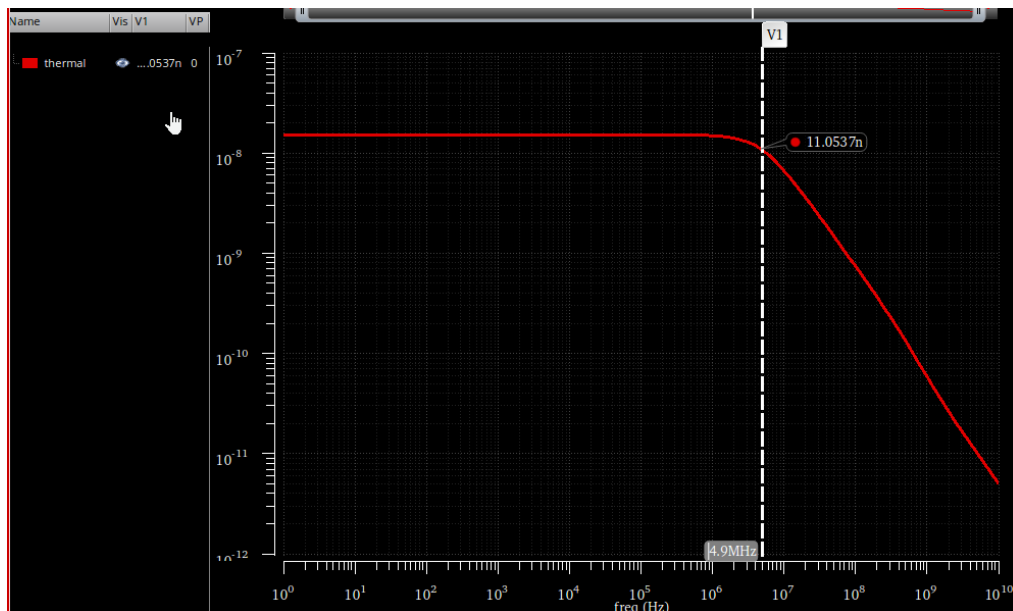


Figure 11: Thermal noise

Thermal Voltage Noise Density: $15.62 \text{ nV}/\sqrt{\text{Hz}}$

Closed-Loop Bandwidth: 4.896 MHz

3.2 Total Output Noise & Flicker Corner

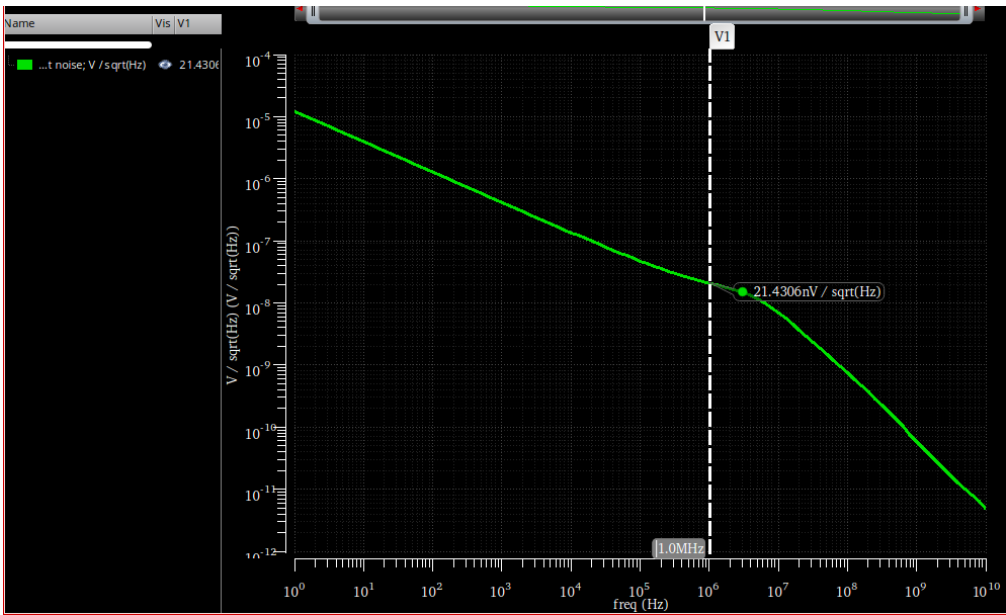


Figure 12: Total output noise spectral density in log scale

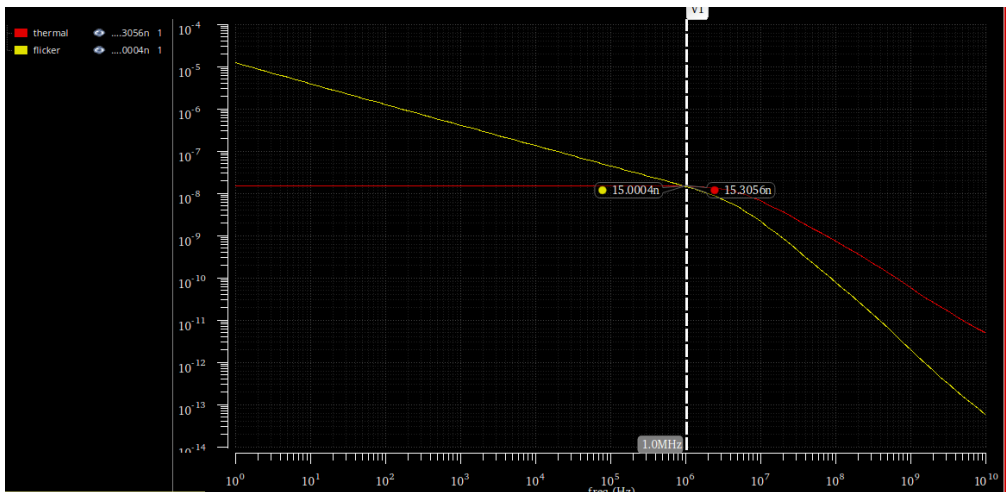


Figure 13: corner frequency

Flicker Noise Corner Frequency (f_c): 1.0 MHz

ITI_LABS:lab_10_noise_ota:1	output noise; V / sqrt(Hz)	
ITI_LABS:lab_10_noise_ota:1	rmsNoise(1 1e+10)	69.85u
ITI_LABS:lab_10_noise_ota:1	sqrt((pv("I0.M1.m0" "id" ?result "...	
ITI_LABS:lab_10_noise_ota:1	flicker	
ITI_LABS:lab_10_noise_ota:1	thermal	
ITI_LABS:lab_10_noise_ota:1	BW	4.896M
ITI_LABS:lab_10_noise_ota:1	thermal power	
ITI_LABS:lab_10_noise_ota:1	flicker power	
ITI_LABS:lab_10_noise_ota:1	dc thermal value	15.62n

Figure 13: the total RMS noise (69.85 μV), bandwidth (4.896 MHz), and DC thermal noise floor (15.62 $nV/\sqrt{Hz}$).

Device	Param	Noise Contribution	% Of Total
I0.M2.m0	fn	1.34555e-09	27.58
I0.M1.m0	fn	1.3021e-09	26.69
I0.M1.m0	id	5.81702e-10	11.92
I0.M2.m0	id	5.74155e-10	11.77
I0.M4.m0	id	3.8814e-10	7.95
I0.M3.m0	id	3.39404e-10	6.96
I0.M3.m0	fn	1.75057e-10	3.59
I0.M4.m0	fn	1.66514e-10	3.41
I0.M6.m0	id	2.87424e-12	0.06
I0.M5.m0	id	2.4786e-12	0.05
I0.M6.m0	fn	8.49158e-13	0.02
I0.M5.m0	fn	4.08074e-13	0.01
I0.M6.m0	rgbi	0	0.00
I0.M6.m0	rs	0	0.00
I0.M6.m0	rd	0	0.00

Integrated Noise Summary (in V²) Sorted By Noise Contributors
Total Summarized Noise = 4.87924e-09
Total Input Referred Noise = 2.04083e-05
The above noise summary info is for noise data with Design_Point = 1.0

Figure 14: Integrated noise summary

Total Simulated Thermal Noise Power = $1.8887 \times 10^{-9} \text{ V}^2$

Simulated RMS Thermal Noise = $\sqrt{1.8887 \times 10^{-9}} = 43.46 \text{ } \mu\text{V}$

$C_L = 5 \text{ pF}$, $g_{m1,2} = 157 \text{ } \mu\text{A/V}$, and $g_{m3,4} = 100 \text{ } \mu\text{A/V}$. assume $\gamma \approx 2/3$.

$$f_{3dB} \approx \frac{g_{m1}}{2\pi C_L}$$

$$f_{3dB} \approx \frac{157 \times 10^{-6}}{2\pi(5 \times 10^{-12})} \approx 4.99 \text{ MHz}$$

$$v_{n,th} = \sqrt{\frac{16kT}{3g_{m1}} \left(1 + \frac{g_{m3}}{g_{m1}}\right)}$$

$$v_{n,th} = \sqrt{\frac{16(1.38 \times 10^{-23})(300)}{3(157 \times 10^{-6})} \left(1 + \frac{100}{157}\right)} \approx 15.16 \text{ nV}/\sqrt{\text{Hz}}$$

$$v_{n,rms,th} = \sqrt{v_{n,th}^2 \times NBW} = \sqrt{\frac{4kT}{3C_L} \left(1 + \frac{g_{m3}}{g_{m1}}\right)}$$

$$v_{n,rms,th} = \sqrt{\frac{4(1.38 \times 10^{-23})(300)}{3(5 \times 10^{-12})} \left(1 + \frac{100}{157}\right)} \approx 42.5 \text{ } \mu\text{V}$$

Parameter	Hand Calculation	Simulation Result
Bandwidth	4.99 MHz	4.896 MHz
Thermal Noise Density	15.16 nV/ $\sqrt{\text{Hz}}$	15.62 nV/ $\sqrt{\text{Hz}}$
Total RMS Noise	42.5 μV (Thermal Only)	43.46 μV

4 PART 4: 5T OTA Transient Noise Analysis

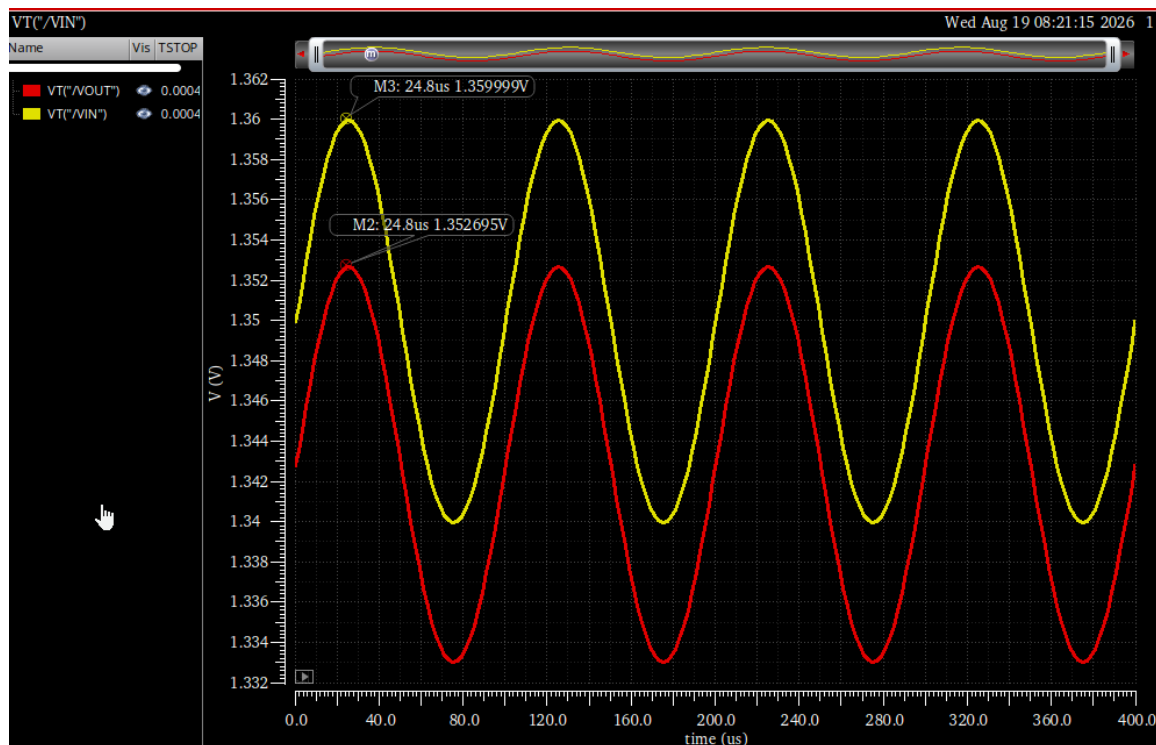


Figure 15: 10 kHz, 10 mV amplitude input signal (yellow) and the resulting output signal (red) with a slight systematic DC offset

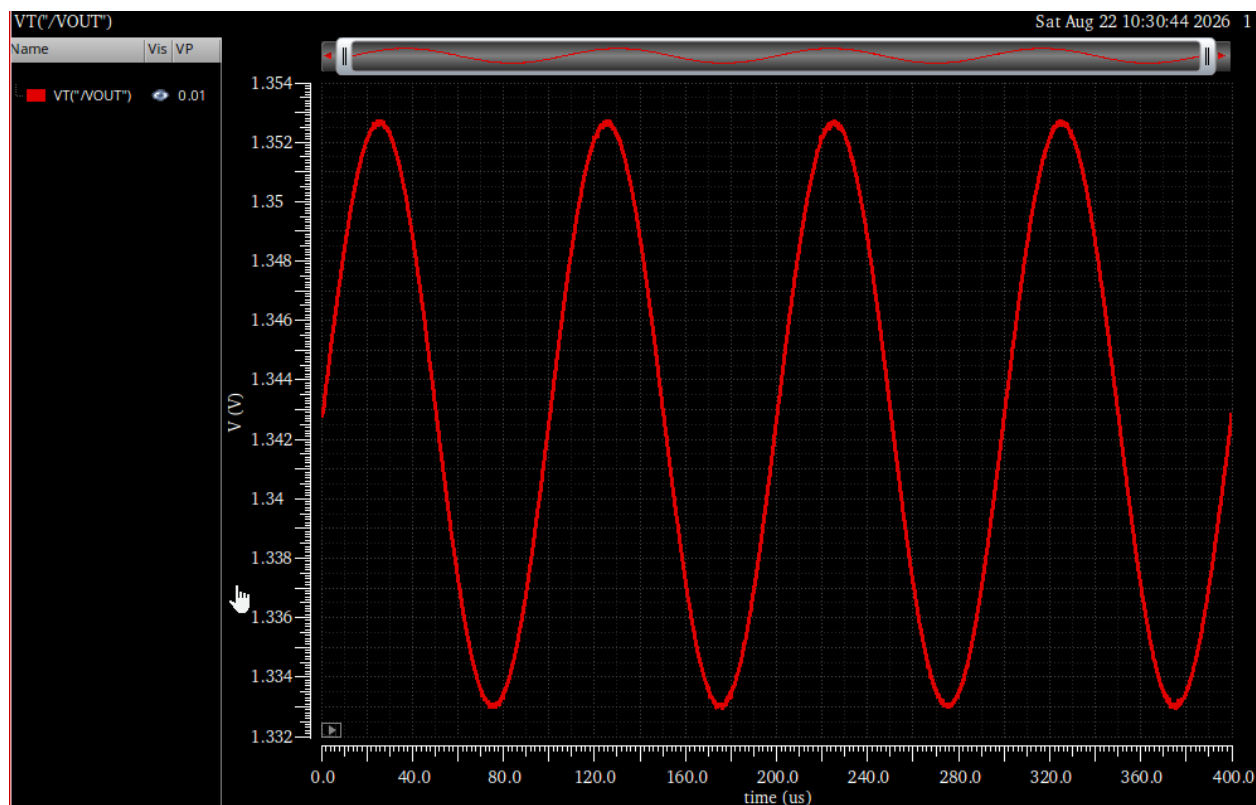


Figure 16: Full transient noisy output waveform

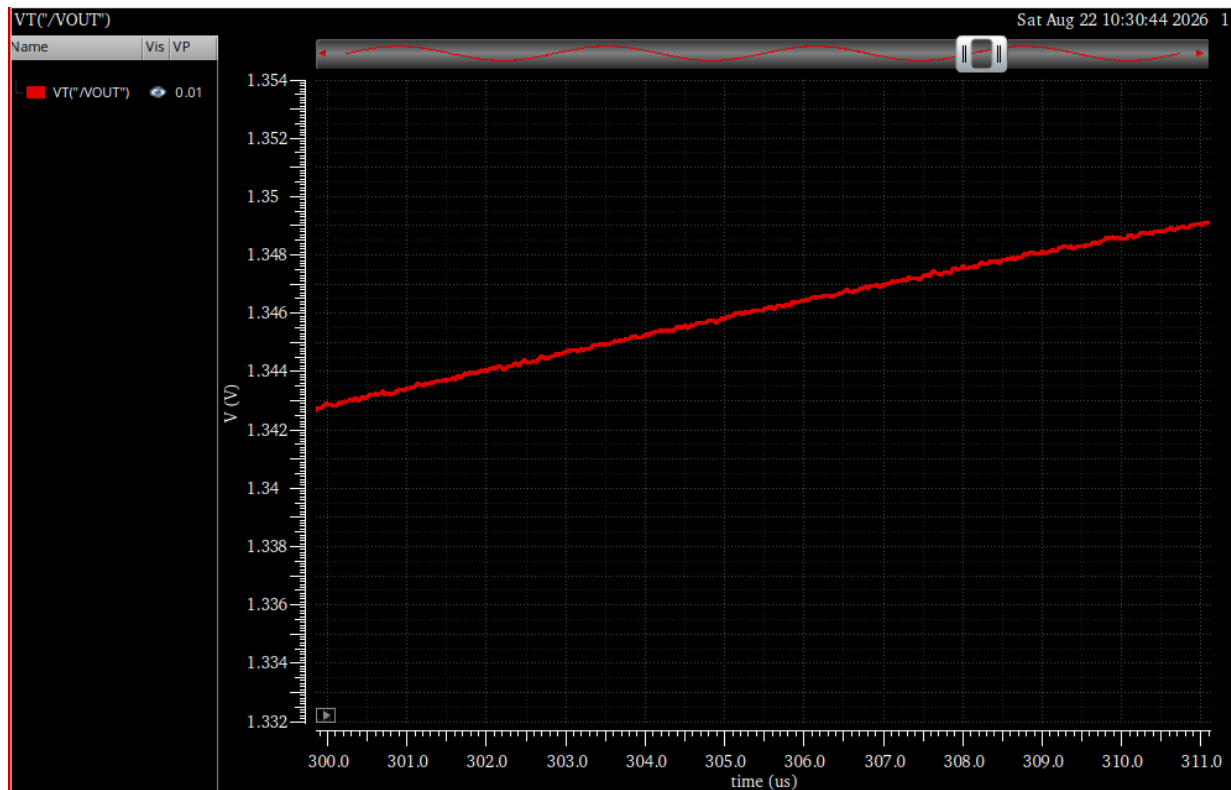


Figure 17: Zoomed-in view of the transient output



Figure 18: RMS vs iterations

From part 3 the thermal rms

$$v_{n,rms,th} \approx 42.5 \mu V$$

Here the average is

$$v_{n,avg} \approx 41.5 \mu V$$

Very close to each other as expected.